python code

- find mean value for $\frac{H}{r}$ using "disk_pop" folder
  ↳ find it for both $\alpha = 1e^{-3}$ and $1e^{-4}$

  ⇒ we will have ONE value for $\frac{H}{r}$ to use in the fortran code

python code

- find five quadratic fits (or polynomial with power > 1), on for each μ (table C.1. left column) for

  $\mathcal{R}$ vs b and $\mathcal{R}$ vs c

  ↑ or $\log_{10}(\mathcal{R})$ ↗

  ⇒ five + five polynomial coefficients to use into the formula of

  $$b = a_0 + a_1 \mathcal{R} + a_2 \mathcal{R}^2 + \dots$$
  $$c = b_0 + b_1 \mathcal{R} + b_2 \mathcal{R}^2 + \dots$$

  and put them in the Fortran code

  (+) do a plot of $\mathcal{R}$ vs b and $\mathcal{R}$ vs c to show the fits over the data points

- write the polynomial fits for b and c into the fortran code "binary_to_modify.f90" and use them to compute R_t as in eq. (C.3) still in the fortran code. Remember:
  - "if" conditions for μ
  - "if" conditions not to go outside the $\mathcal{R}$ boundaries which are $10^4 < \mathcal{R} < 10^6$

PROJECT IS DONE

- to check if R_t makes sense (for example R_t for $a_{bin} = 100$ au is greater than $a_{bin} = 50$ au for equal other parameters) you can write a separate python code like this:

$M_1 = \ldots$

$M_2 = \ldots$

$\alpha = \ldots$

$e = \ldots$

$a_{bin} = \ldots$

ADD formulas to compute R_t

print (R_t)